

CORENTIN PUFFAY

Post-doctoral researcher at KU Leuven - EPFL graduate

@ corentin.puffay@kuleuven.be +33658898342 Vaartkom 5C 01.04, 3000 Leuven, Belgium
in linkedin.com/in/corentin-puffay-389863151 https://corentinpuffay.github.io/



SUMMARY

I am a researcher fascinated by neuroscience, language, and deep learning. At EPFL, I developed a strong engineering foundation through coursework in programming, machine learning, and neuroscience. My Ph.D. at KU Leuven focused on relating continuous speech to brain signals (EEG) using deep learning architectures, to model how the brain processes speech. In parallel, I explored language representations, leveraging e.g., large language models (LLMs), to probe how the brain encodes language across multiple modalities, including single-unit recordings, sEEG, ECoG, and EEG. My broad research interests are centered on understanding and modeling how the brain transforms sounds into meaning using novel deep learning models, for both fundamental research and translational applications.

EDUCATION

Postdoctoral researcher in speech NeuroAI - ExpORL/PSI

Katholieke Universiteit Leuven (KU Leuven)

June 2025 – Now Leuven, Belgium

- Modeling language context encoding in the brain using LLMs, including effects of auditory attention.
- Investigating the dynamics of hierarchical language processing (phonetic, syntactic, semantic) in the hippocampus.
- Designing EEG experiments to probe frequency-specific brain responses for hearing assessment in naturalistic settings.

PhD in Biomedical Sciences and Engineering Science:
Electrical Engineering - ExpORL/PSI groups

Katholieke Universiteit Leuven (KU Leuven)

Feb. 2021 – June 2025 Leuven, Belgium

- Developed deep learning architectures to relate EEG recordings to natural speech.
- Modeled the auditory system using features from acoustics to semantics, capturing brain responses across the speech processing hierarchy.

MSc in Bioengineering

Swiss Federal Institute Of Technology (EPFL)

Feb. 2018 – Mar. 2020 Lausanne, Switzerland

- Specialization in Neuroprosthetics
- Average grade 5.1/6

BSc in Life Science Engineering

Swiss Federal Institute Of Technology (EPFL)

Sep. 2013 – Dec. 2017 Lausanne, Switzerland

SKILLS

Python Tensorflow Matlab
HuggingFace LLMs C++ R LaTeX

LANGUAGES

French ●●●●●
Mother tongue

English ●●●●●
Fluent

Dutch ●●●●●
Intermediate

Spanish ●●●●●
Intermediate

German ●●●●●
Basic

TEACHING

Teaching Assistant
Swiss Federal Institute of Technology (EPFL)
Sep. 2014– Jan. 2016 Lausanne, Switzerland

- Management of exercise sessions with student groups for General Chemistry.

EXTRA-CURRICULAR

- Crossfit, Running, Guitar Playing, Literature

RESEARCH EXPERIENCE

Visiting researcher - Gwilliams Laboratory of Speech Neuroscience (Prof. Laura Gwilliams)

Stanford University

📅 March 2026-July 2026

📍 Stanford, United States

- Modeling the dynamics of language hierarchy encoding in hippocampus single units recording.

Visiting researcher - Neural Acoustic Processing Laboratory (Prof. Nima Mesgarani)

Columbia University

📅 June 2024-October 2024

📍 New-York City (NY), United States

- Model the impact of auditory attention on context encoding in intracranial brain responses using Large Language Models.

Master Thesis - Brown Neuromotion Laboratory

Brown University

📅 Sep. 2019 – Mar. 2020

📍 Providence (RI), United States

- Modeling the effects of spinal cord stimulation on the somatosensory cortex response measured intracranially in Rhesus monkeys.

Semester Project - Prof. Courtine Laboratory

Center for Neuroprosthetics, EPFL

📅 Feb. 2019 – Jun. 2019

📍 Lausanne, Switzerland

- Modeling spinal cord stimulations using finite element models analyses to restore locomotion in spinal cord injury patients

ACADEMIC GRANTS

- **PhD fellowship strategic basic research (October 2022)** - Research Foundation Flanders (FWO)
Project: Diagnostics of the auditory system using deep-learning-based analysis of EEG signals.
- **Travel grant for long stay abroad (June 2024, Columbia University)** - Research Foundation Flanders (FWO)
Project: Modeling language processing in intracranial recordings using large language models.
- **Travel grant for long stay abroad (March 2026, Stanford University)** - Research Foundation Flanders (FWO)
Project: Modeling the dynamics of language hierarchy encoding in hippocampus single units recording.

PUBLICATIONS

- **C. Puffay**, G. Mischler, V. Choudhari, J. Vanthornhout, S. Bickel, A. D. Mehta, C. Schevon, G. M. McKhann, H. Van hamme, T. Francart, N. Mesgarani Large Language Models Reveal the Neural Tracking of Linguistic Context in Attended and Unattended Multi-Talker Speech. *Imaging Neuroscience* (2026).
- He, L., Nie, E., Dindar, S. S., Firoozi, A., Florea, A., Nguyen, V., **Puffay, C.**, Shimizu, R., Ye, H., Brennan, J., Schmid, H., Schütze, H., and Mesgarani, N. (2025). XCOMPS: A multilingual benchmark of conceptual minimal pairs. *In Proceedings of the 7th Workshop on Research in Computational Linguistic Typology and Multilingual NLP, pages 75–81, Vienna, Austria. Association for Computational Linguistics.*
- **C. Puffay**, J. Vanthornhout, M. Gillis, P. De Clercq, B. Accou, H. Van hamme, T. Francart. Classifying coherent versus nonsense speech perception from EEG using linguistic speech features. *Scientific Reports* 14, 18922 (2024).

SOCIAL INITIATIVES

Hackahealth 2019 – Designed assistive devices to address daily challenges faced by people with disabilities.

DISTINCTIONS

Winner NeurIPS EEG Foundation Challenge

📅 2025

📍 San Diego, CA, USA

- First place in an international EEG decoding competition using deep neural network architectures.

1st prize - French Physics Olympiads

📅 2011 – 2012

📍 Paris, France

- Realisation of a collaborative geophysics project, presented to a jury of physicists in Paris.

REFEREES

- Prof. Tom Francart - KU Leuven (tom.francart@kuleuven.be)
- Prof. Hugo Van hamme - KU Leuven (hugo.vanhamme@esat.kuleuven.be)
- Prof. Nima Mesgarani - Columbia University (nima@ee.columbia.edu)
- Prof. Laura Gwilliams - Stanford University (laura.gwilliams@stanford.edu)

- P. De Clercq, **C. Puffay**, J. Kries, H. Van Hamme, M. Vandermosten, T. Francart, J. Vanthornhout, (2024). Detecting Post-Stroke Aphasia Via Brain Responses to Speech in a Deep Learning Framework. *IEEE Engineering in Medicine and Biology Conference (EMBC 2024)*
- L. Bollens, **C. Puffay**, B. Accou, J. Vanthornhout, H. Van Hamme, T. Francart, (2024). Auditory EEG decoding challenge for ICASSP 2024. *IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP) 2024*.
- **C. Puffay**, J. Vanthornhout, M. Gillis, B. Accou H. Van hamme, T. Francart, (2023). Robust neural tracking of linguistic speech representations using a convolutional neural network. *Journal of Neural Engineering, IOP publishing*
- **C. Puffay**, B. Accou, L. Bollens, M.J. Monesi, J. Vanthornhout, H. Van hamme, T. Francart, (2023). Relating EEG to continuous speech using deep neural networks: a review, *Journal of Neural Engineering, IOP publishing*
- **C. Puffay**, J. Van Canneyt, J. Vanthornhout, H. Van hamme, T. Francart, (2022). Relating the fundamental frequency of speech using a dilated convolutional network, *Proc. Interspeech 2022 10.21437/Interspeech.2022-315*